

Electric Bill

Title: To calculate Electric bill

Topics: Range, In between, Integer, float, input functions, variables

Problem Statement:

To design and implement a program that calculate the electricity bill based on the number of units consumed using slab-wise traffic rates and fixed service charges ensuring accurate and automated billing

Roles& Responsibilities:

Project Manager:

- o Team coordination
- o Final project to delivery client

Backend Developer:

- o Billing logical implement
- o Correct output to generate

Frontend Developer:

- o To create a form entering units
- o To display output

Tester:

- o Identifying bugs
- o To check the bill calculate

Code:

Using range :

```
def calculate_bill(units):  
    minimum_price = 200    # Minimum charge for first  
    50 units  
    service_charge = 50    # Fixed service charge  
    bill = 0  
  
    if units in range(0, 51):        # 0–50 units  
        bill = minimum_price  
  
    elif units in range(51, 101):     # 51–100 units  
        bill = minimum_price + (units - 50) * 5  
  
    elif units in range(101, 201):    # 101–200 units  
        bill = minimum_price + (50 * 5) + (units - 100) * 8  
  
    elif units in range(201, 301):    # 201–300 units  
        bill = minimum_price + (50 * 5) + (100 * 8) + (units -  
200) * 3
```

```
else:                                # Above 300 units
```

```
    bill = (  
        minimum_price +  
        (50 * 5) +  
        (100 * 8) +  
        (100 * 3) +  
        (units - 300) * 22  
    )
```

```
return bill + service_charge
```

```
# ----- DECLARED VARIABLE (NO INPUT) -----
```

```
units = 120      # you can change this value
```

```
amount = calculate_bill(units)
```

```
print("Units:", units)
```

```
print("Your bill:", amount)
```

Using In between:

```
def calculate_bill(units):  
    service_charge = 50  
    bill = 0  
  
    if 0 <= units <= 50:  
        bill = 200  
    elif units <= 100:  
        bill = 50 * 5 + (units - 50) * 5  
    elif units <= 200:  
        bill = (50 * 5) + (50 * 5) + (units - 100) * 8  
    elif units <= 300:  
        bill = (50 * 5) + (50 * 5) + (100 * 8) + (units - 200) * 13  
    else:  
        bill = (50 * 5) + (50 * 5) + (100 * 8) + (100 * 13) + (units -  
300) * 22  
  
    return bill + service_charge
```

```
units = float(input("Enter units: "))
```

```
amount = calculate_bill(units)
```

```
print("Your bill:", amount)
```

Using Variables(Int):

```
def calculate_bill(units):
```

```
    service_charge = 50
```

```
    bill = 0
```

```
    if 0 <= units <= 50:
```

```
        bill = 200 # minimum price
```

```
    elif 51 <= units <= 100:
```

```
        bill = 50 * 5 + (units - 50) * 5
```

```
    elif 101 <= units <= 200:
```

```
        bill = (50 * 5) + (50 * 5) + (units - 100) * 8
```

```
    elif 201 <= units <= 300:
```

```
bill = (50 * 5) + (50 * 5) + (100 * 8) + (units - 200) * 13
```

```
else:
```

```
    bill = (50 * 5) + (50 * 5) + (100 * 8) + (100 * 13) + (units -  
300) * 22
```

```
    return bill + service_charge
```

```
# using int variable
```

```
units = int(input("Enter your units: "))
```

```
amount = calculate_bill(units)
```

```
print("Your bill:", amount)
```

```
    Using float (variable):
```

```
def calculate_bill(units):
```

```
    minimum_price = 200
```

```
    service_charge = 50
```

```
if 0 <= units <= 50:
```

```
    bill = minimum_price
```

```
elif units <= 100:
```

```
    bill = minimum_price + (units - 50) * 5
```

```
elif units <= 200:
```

```
    bill = minimum_price + (50 * 5) + (units - 100) * 8
```

```
elif units <= 300:
```

```
    bill = minimum_price + (50 * 5) + (100 * 8) + (units - 200) *
```

```
3
```

```
else:
```

```
    bill = (
```

```
        minimum_price +
```

```
        (50 * 5) +
```

```
        (100 * 8) +
```

```
        (100 * 3) +
```

```
(units - 300) * 22  
)
```

```
return bill + service_charge
```

```
# ----- INPUT & OUTPUT -----
```

```
units = int(float(input("Enter your units: "))) #    main logic
```

```
amount = calculate_bill(units)
```

```
print("Units:", units)
```

```
print("Your bill:", amount)
```

Output :

✓ Using Range:

- o Units:250
- o Your bill:2000

✓ Using in between:

- o Units:250

o Your bill:2000

✓ Using variable(int):

o Units :200

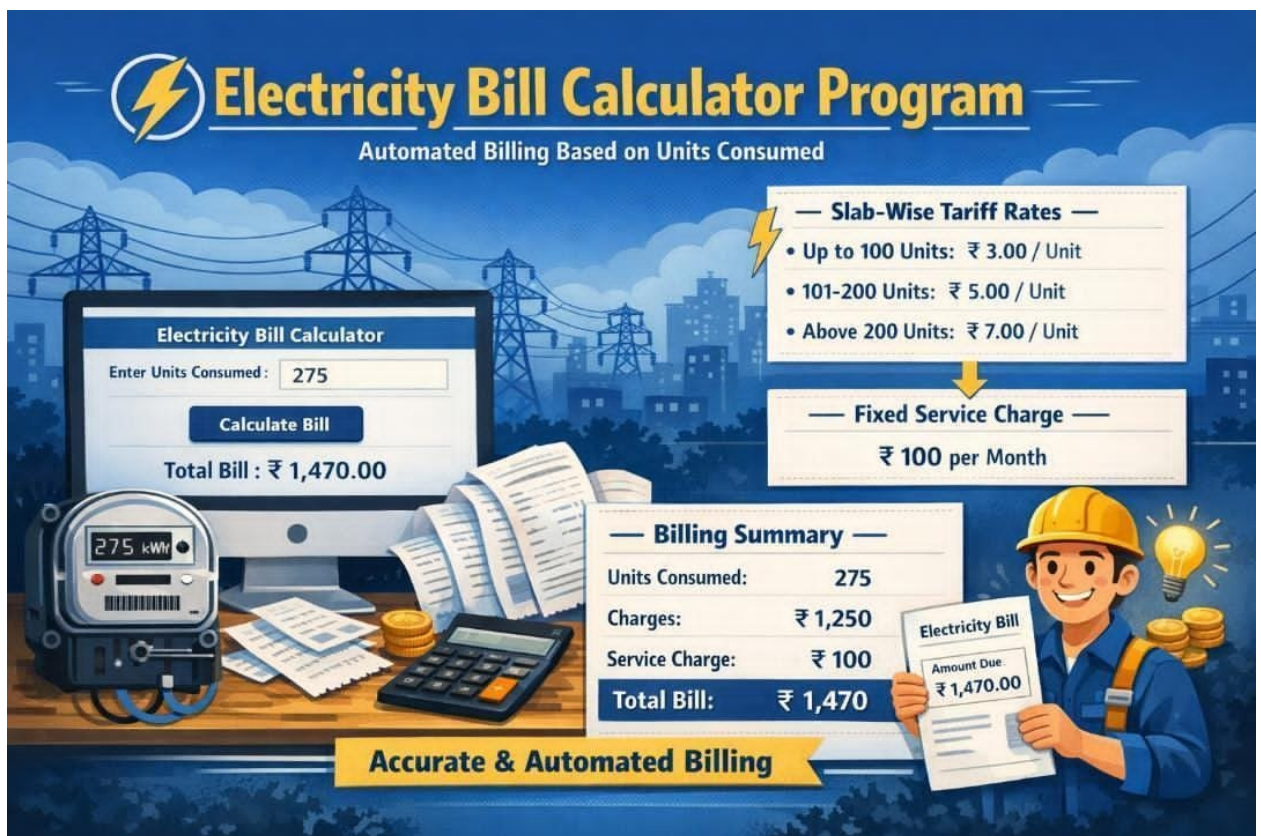
o Your bill:1300

✓ Using float:

o Units:120.5

o Your bill:660

Image:



Interview Questions:

- ✓ What is the aim of the project
 - o Automation
- ✓ What does the program calculate
 - o Electricity bill
- ✓ What is the input/output of the program
 - o Units/amount
- ✓ Which logic is used for calculate
 - o Slab
- ✓ Which programming concept is mainly used
 - o Conditional Statements
- ✓ Why are function used
 - o Reusability
- ✓ What is the main advantages of this system
 - o Efficiency/Time Consumption